Two Varieties, Two Sarcasms: A Validated Ensemble and a Close Look at Australian and British Sarcasm Labels

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Abstract

We describe our system for the ALTA 2026 shared task, which asks for sentiment and sarcasm labels on Australian (en-AU) and British (en-UK) English Google reviews and Reddit comments, and scores each task by its weaker variety. An ensemble of a fine-tuned DeBERTa-v3-large and two linear probes, chosen under a strict validation protocol, scored 0.867 on the test set: second on the leaderboard, and one of six teams the organisers declared joint winners. We then examine where so many different models stall. For our two strongest members, going from half to all of the training data helps little, as does going from three to ten seeds, and the two varieties' sarcasm labels turn out to differ markedly. Among Reddit comments that a sentiment model calls negative, every en-AU subreddit has a higher sarcastic share (44–64%) than every en-UK one (12–33%), and derision cues (mocking, contempt, anger) predict an en-AU sarcasm label, while contempt and anger predict the absence of an en-UK one and playful cues predict en-UK sarcasm more. Both patterns are compatible with the task guideline, which counts "irony or mockery". With one annotator per variety we cannot tell whether they stem from the annotators, the communities or the texts, but they are consistent with the poor cross-variety transfer of sarcasm and with where our binding en-AU errors lie: deciding which negative comments count as sarcastic.

1 Introduction

The ALTA 2026 shared task (Mollá-Aliod et al., 2026) uses the Australian (en-AU) and British (en-UK) English portions of BESSTIE (Srirag et al., 2025). Each text is a Google review or a Reddit comment with a binary sentiment label and a binary sarcasm label. Each task is scored by macro-F1, computed separately for each variety. The official score keeps the lower of the two varieties for each task and averages the two tasks. We call the variety with the lower score the *binding* variety, since only its score counts. For example, our test scores of 0.932 (en-AU) and 0.976 (en-UK) for sentiment and 0.801 and 0.837 for sarcasm give $(0.932 + 0.801)/2 \approx 0.867$.

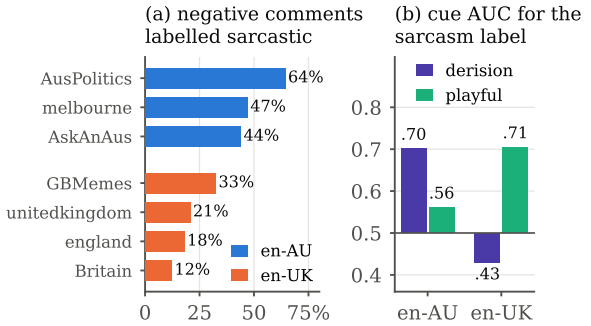


Figure 1: The two varieties' sarcasm labels differ. (a) Among Reddit comments that an off-the-shelf sentiment model (SiEBERT) rates negative, the share that the dataset's annotator labelled sarcastic, per subreddit (archive-matched comments: 1,221 en-AU, 883 en-UK). (b) How well four derision cues and four playful cues predict the annotator's sarcasm label, as mean AUC (0.5 = no better than chance); the cues are scores from off-the-shelf classifiers that never saw this task's labels (details in Figure 4). Neither panel uses the LLM re-annotation of Appendix B.

This paper makes two points. First, a small ensemble chosen under a strict protocol was enough to share first place. Second, and more useful for anyone working with this data, the en-AU and en-UK sarcasm labels differ markedly in rate and in the cues they follow. This matters for how the benchmark is used: when a model trained on one variety fails on the other, part of the failure may reflect different labelling conventions rather than a failure to understand the variety. Figure 1 summarises the evidence: the en-AU labels call far more negative comments sarcastic, in every subreddit, and contempt and anger, which go with an en-AU sarcasm label, go with its absence in en-UK. These analyses locate difficult cases and suggest hypotheses; they do not identify a single cause of

the plateau.

Our findings, in order:

1. A three-model ensemble scored 0.867 on the test set, second on the leaderboard and one of six teams the organisers declared joint winners (§5).
2. For DeBERTa and the LLM probe, going from half to all of the training data helps little, as does going from three to ten seeds; the gains come from combining different model families (§6.1).
3. The two varieties' sarcasm labels differ in rate and in which cues they follow: derision predicts an en-AU sarcasm label, while contempt and anger predict the absence of an en-UK one in every subreddit (§6.2).
4. Agreement on en-AU sarcasm is low even between humans, and our ensemble is weakest at telling which negative en-AU comments are labelled sarcastic (§6.3).

We also report what did not help (§7) and, as an exploratory check, a blind re-annotation by an LLM (Appendix B).

Related work. Two findings on sarcasm bear directly on ours: Oprea and Magdy (2020a) show that intended and perceived sarcasm differ, and Oprea and Magdy (2020b) report differences in how sarcasm is interpreted associated with country, age and whether English is the reader's native language. BESSTIE extends variety benchmarks such as DIALECTBENCH (Faisal et al., 2024) to sentiment and sarcasm in English varieties, and its authors already reported that sarcasm transfers poorly across varieties (Srirag et al., 2025); Singh et al. (2025) add explanations to BESSTIE's Australian and Indian English sarcasm data and evaluate pragmatic metacognitive prompting. Our analysis belongs to the literature on label variation and annotator effects (Geva et al., 2019; Basile et al., 2021; Plank, 2022): Röttger et al. (2022) distinguish descriptive annotation, which elicits annotators' own judgements, from prescriptive annotation, which asks them to apply one specified interpretation, and the EPIC corpus found and modelled systematic differences between irony annotators from five English-speaking countries, including the UK and Australia (Frenda et al., 2023). On the linguistic side, jocular mockery and abuse, which combine apparent hostility with affiliation, have been described in both Australian and British interaction (Haugh and Bousfield, 2012). Our contribution is narrower and benchmark-specific: a competitive system for this shared task, and a within-subreddit analysis of how the two BESSTIE sarcasm label sets relate to label-free cues.

2 Task and Data

The training and development files together (the *pool*) hold 6,288 texts: 3,080 en-AU (1,076 reviews, 2,004 Reddit comments) and 3,208 en-UK (2,065 reviews, 1,143 comments). We call each variety–source combination a *segment*. Sarcasm lives almost entirely on Reddit: 42% of en-AU and 22% of en-UK comments are labelled sarcastic, against 6.8% of en-AU reviews and a single en-UK review. The test set has 1,573 texts in the same proportions.

BESSTIE selected Reddit comments from variety-specific subreddits and Google reviews of places in each country, so the variety tags are dataset assignments, not verified speaker identities; texts without clear polarity were discarded. It used one annotator per variety. Its guideline reads: "Annotate if the comment uses irony or mockery to express contempt or ridicule," and asks annotators to "consider the tone and context of the comment" (Srirag et al., 2025). The guideline thus admits two readings, an inclusive one in which mockery alone counts and a narrower one centred on irony. Table 1 illustrates both readings in the data.

The official split is by comment, not by Reddit thread. By matching Reddit texts to a public comment archive (Heitmann, 2024) we recovered 95% of them: they come from seven subreddits between February and June 2024, and 82% of the matched development comments share a thread with a training comment. Re-running an ensemble with three DeBERTa seeds on thread-grouped folds lowers its score by only 0.001 (0.871 to 0.869); only the e5 probe loses more (0.008). Within this audit, the comment-level split does not materially inflate our estimates.

3 Validation Protocol

With small, noisy data and a metric that takes the weaker variety, most apparent improvements are noise. We fixed our evaluation early in development, before any model was selected. Each safeguard answers one risk.

- **Overfitting the development file.** We pool training and development data and use five folds, stratified by segment and labels and keep-

	Comment (paraphrased)	gold	ours
Derision labelled sarcastic (en-AU)			
AU	A private-school boy playing gangster. Couldn't be more embarrassing if he tried.	S	S
AU	Couldn't have happened to a more deserving creep. Good riddance.	S	S
Irony labelled sarcastic (en-UK)			
UK	A sensible policy from *that* party? Has Hell frozen over?	S	S
UK	I, for one, welcome our new pigeon overlords.	S	S
Hostility labelled not sarcastic (en-UK)			
UK	How did this clown end up running the country? Total knuckle-dragger.	–	–
UK	Can we stop giving this con artist airtime? Half the country already fell for him once.	–	–

Table 1: Examples selected to illustrate the two readings of the guideline (not a random sample); real comments, paraphrased. S = sarcastic, – = not sarcastic; "ours" is our ensemble's label. The comments are users' words about public figures.

ing identical texts together. Members and ensembles are judged on out-of-fold (OOF) predictions for all 6,288 texts; the official development file serves only as a check.

- **Optimistic thresholds.** We *cross-fit* decision thresholds: one threshold per task and segment is fitted on four folds and applied to the fifth, so reported scores include the cost of choosing it. For the test set, thresholds are fitted on all OOF predictions and applied to the average of the five fold models.
- **Mistaking noise for gains.** We compare two candidates with a paired bootstrap (Efron, 1979; Koehn, 2004): 2,000 test-sized resamples of the pool, stratified by segment and labels, with predictions and thresholds held fixed. We report P_{boot}, the share of resamples in which one beats the other. It covers test sampling only, not retraining or selection, and it is not a p-value: two seeds of the same MLP probe reach $P_{\text{boot}} \geq 0.9$ in 18% of pairs. We call $P_{\text{boot}} < 0.8$ no evidence and $0.8 \leq P_{\text{boot}} < 0.95$ suggestive. Because fine-tuning results vary with the random seed (Dodge et al., 2020), we compare neural models by averages over at least two seeds.
- **Leakage in our pipeline.** Models are re-run with shuffled labels and must fall to the no-information score; all three members of the final system do (0.63–0.64, against a floor of 0.64).

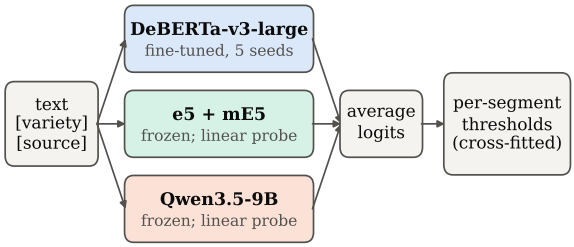


Figure 2: The submitted system. Three models score each text; their logits are averaged with equal weights, and a threshold for each task and segment, learned from out-of-fold predictions, turns the average into labels.

- **Selection on one lucky split.** System rankings can depend on the split (Gorman and Bedrick, 2019). Neural recipes are therefore screened on one fold, and anything promising is confirmed on an untouched fold, with the success criterion written down beforehand in the spirit of preregistration (Nosek et al., 2018); these plans are internal documents, not entries in a public registry.

4 System

The system (Figure 2) averages three models that see the data in different ways.

Fine-tuned encoder. DeBERTa-v3-large (He et al., 2023), fine-tuned for four epochs (learning rate 10^{-5}, batch size 16). The input is prefixed with its variety and source, e.g. `[en-UK] [reddit]`. One output predicts sentiment; sarcasm has a separate output for each variety. We train five seeds on each of the five folds and average the 25 models on the test set.

Embedding probe. A logistic regression (Pedregosa et al., 2011) on frozen sentence embeddings from e5-large-v2 (Wang et al., 2022) and multilingual-e5-large-instruct (Wang et al., 2024), with regularisation chosen by inner cross-validation.

LLM probe. Qwen3.5-9B (Qwen Team, 2026), quantised to 4 bits (Dettmers et al., 2023), reads each text inside a zero-shot sarcasm (or sentiment) prompt; the sarcasm prompt's definition differs by variety (Appendix C). We train a logistic regression, i.e. a linear probe (Alain and Bengio, 2017), on two inputs: the model's hidden state at layer 24, and the difference between the logits of the "yes" and "no" answer words. Each variety gets its own sarcasm probe.

	Sentiment		Sarcasm		
	AU	UK	AU	UK	Score
DeBERTa ×5, OOF	.931	.972	.803	.816	.867
e5 probe, OOF	.917	.969	.777	.788	.847
9B LLM probe, OOF	.916	.973	.789	.806	.853
Ensemble, OOF	.934	.976	.805	.841	.869
Ensemble, **test**	.932	.976	.801	.837	**.867**
DeBERTa ×5, **test**	.916	.975	.793	.820	.854
Organisers' baseline, test	.866	.931	.680	.696	.773

Table 2: Macro-F1 per component; score $= \frac{1}{2}[\min(\text{sent})+\min(\text{sarc})]$. OOF: out-of-fold predictions on the pool, with cross-fitted thresholds. The baseline is the organisers' fine-tuned DistilBERT (Sanh et al., 2019; Srirag, 2026); its test scores are from the official leaderboard.

Combination. We average the three models' logits with equal weights. Learned weights and stacking (Wolpert, 1992) did not do better (§7). Because en-UK reviews contain a single sarcastic example, we label all of them not sarcastic.

5 Results

The test score fell inside the predicted range. The ensemble scored 0.8668 on the test set (Table 2), 0.002 below its OOF estimate and inside the heuristic 90% band we computed before submitting (0.844–0.895: the OOF score ± 1.645 bootstrap standard deviations, widened by 20% as a margin for selection, as our red-team review recommended). No component moved by more than 0.004. En-AU had the lower F1 in both tasks, so en-AU set the official score. With a sampling standard deviation of about 0.013 (from resampling pool predictions at test size), the closeness is partly luck.

It shares first place. Our run placed second on the leaderboard, 0.0007 behind the top run. After an approximate randomisation test with 10,000 swaps (Dror et al., 2018) did not separate the top six teams, the organisers declared all six joint winners. A non-significant difference does not show that the runs are equivalent; it shows that this test set cannot rank them. According to the organisers, our run and the top run disagree on 2.1% of sentiment and 4.7% of sarcasm labels among the 1,573 test texts.

The probes helped on test. As an ablation we also submitted the DeBERTa models alone. They scored 0.854, against 0.867 for the ensemble. The paired bootstrap on the pool had predicted a gain of +0.005 (90% band -0.010 to +0.023), so the observed +0.013 is high but inside the band. On the pool the gap was only 0.002 because these five seeds happen to be strong: averaged over seed subsets, adding the probes is worth about 0.007 (Figure 3b). The test loss came almost entirely from the two binding en-AU components (sentiment 0.931 to 0.916, sarcasm 0.803 to 0.793).

6 Beyond the Score: Data, Seeds and Labels

The competitive models and recipes we tried landed within about 0.01 of each other (§7). This section examines three possible limits in turn; it locates difficult cases rather than identifying a single cause.

6.1 Is it a data or seed limit?

For DeBERTa and the LLM probe, more data beyond half of the pool helps little, and more seeds beyond three help little for DeBERTa (Figure 3).

More data. With matched seeds, DeBERTa scores 0.858, 0.859 and 0.862 at 50, 75 and 100% of the training data, a gain of 0.004; the LLM probe scores 0.853, 0.854 and 0.853. These curves use a fixed number of epochs, so smaller subsets also get fewer updates. Sarcasm AUC still rises slightly in both varieties, and more for en-UK, which has only 248 sarcastic Reddit comments (DeBERTa, 50 to 100%: en-UK 0.845 to 0.867, en-AU 0.833 to 0.845). The e5 probe still gains 0.010 from 50 to 100%. These curves show diminishing returns within the pool; they cannot say what new data or larger models would do.

More seeds. With ten DeBERTa seeds, averaging adds little beyond three (0.860 for one seed, 0.863 for three, 0.864 for ten). The two probes add about 0.007 on top, so the ensemble's gain comes from model diversity rather than averaging. Two five-seed sets of the same recipe differ by 0.004, a useful reference for the size of seed noise.

Design choices. We ablated three choices, averaging two seeds each (baseline 0.865). Dropping the `[variety] [source]` prefix costs 0.008 (suggestive: $P_{\text{boot}} = 0.84$, worse on 4 of 5 folds). Training two single-task models instead of one multi-task model changes nothing (0.865), and sharing one sarcasm output across varieties stays within noise (0.863).

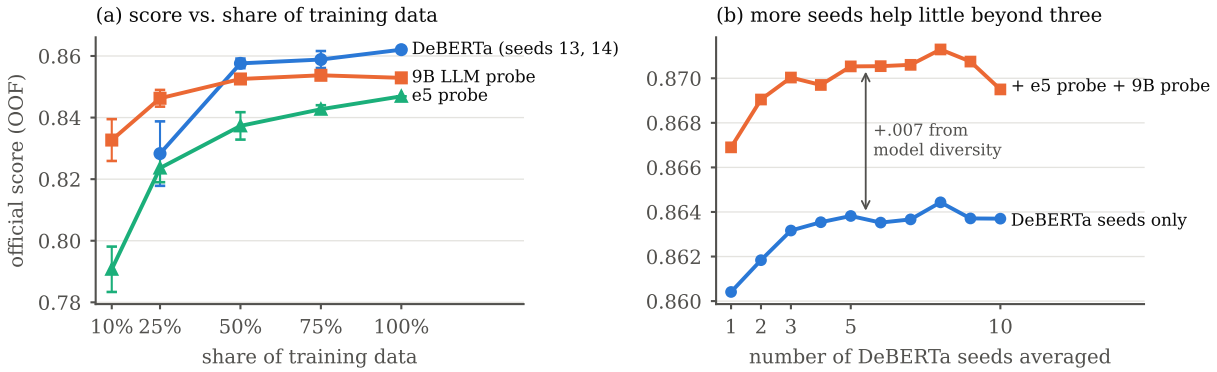


Figure 3: Where gains slow down. (a) OOF score when each model is trained on a nested, segment-stratified share of the training folds (DeBERTa: mean of two seeds; probes: three subset draws below 100%; error bars: standard deviation). (b) Averaging more DeBERTa seeds adds little beyond three (each point averages over sampled subsets of ten seeds); adding the two probes adds about 0.007 at any number of seeds. Note the narrow y-axis: the 0.003 gain from one to three seeds is below the 0.004 difference between two five-seed sets of the same recipe.

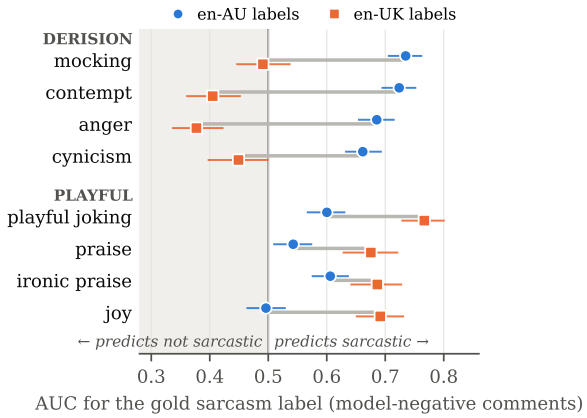


Figure 4: Derision cues predict an en-AU sarcasm label; contempt and anger predict its absence in en-UK. AUC of label-free zero-shot cues for the gold sarcasm label among Reddit comments that a sentiment model (Hartmann et al., 2023) calls negative (en-AU n = 1,271, 685 sarcastic; en-UK n = 912, 186 sarcastic), with pointwise 95% bootstrap intervals. Right of 0.5 a cue predicts a sarcasm label; left of 0.5 it predicts its absence.

6.2 Do the two varieties label the same sarcasm?

The observed label distributions and cue associations differ. In both varieties almost every sarcastic comment is negative (97% en-AU, 99% en-UK), but the en-AU annotator labels 60% of negative Reddit comments sarcastic, against 24% in en-UK. The gap is not driven by a few communities: among comments that a label-free sentiment model (Hartmann et al., 2023) calls negative, every en-AU subreddit (44–64%) has a higher sarcastic share than every en-UK one (12–33%; Figure 1a).

Derision cues behave differently. We scored every comment with label-free zero-shot cues from an NLI classifier (Laurer et al., 2023) and an emotion classifier (Hartmann, 2022), and measured how well each predicts the gold sarcasm label (Figure 4). To avoid conditioning on the annotator's own labels, we restrict to comments that a separate sentiment model calls negative, and also report the unfiltered result. Derision cues predict an en-AU sarcasm label (mocking 0.73, contempt 0.72, anger 0.69, cynicism 0.66). In en-UK, contempt and anger predict its absence (0.41 and 0.38), and mocking and cynicism carry little signal (0.49 and 0.45). Playful cues predict en-UK sarcasm more strongly (joking 0.77, praise 0.68, ironic praise 0.69, joy 0.69) than en-AU sarcasm (0.50–0.61). The same contrast holds over all Reddit comments without any filter (derision 0.76–0.80 for en-AU, 0.40–0.51 for en-UK) and within each subreddit (three en-AU, four en-UK): all four en-AU derision cues score 0.62–0.74 in every en-AU subreddit, and en-UK contempt and anger score 0.33–0.47 in every en-UK subreddit (en-UK mocking varies more, 0.37–0.56). The contrast is therefore not an artefact of mixing communities, although the two varieties' comments still come from different communities. These eight cues were chosen as interpretable representatives of derision and play from 18 NLI hypotheses and an emotion model; they are exploratory.

Two readings of one guideline, three possible causes. BESSTIE's guideline counts "irony or mockery". The en-AU labels behave like its inclusive reading, in which mockery that expresses contempt is sarcasm; the en-UK labels behave like

a narrower reading centred on irony, in which open hostility is not. Three explanations fit this pattern, and our data cannot separate them: the two annotators may apply the guideline differently; the communities may differ in how they use mockery, which can combine apparent hostility with affiliation (Haugh and Bousfield, 2012); or the comments themselves may differ, since each annotator labelled different texts. The cue scores are also classifier outputs whose errors may differ between varieties. A crossed design, in which several native annotators of each variety label the same comments, would help distinguish annotator and item effects. Our own system hints at the same split: during development, the LLM probe's prompt worked best with a broad sarcasm definition that includes derisive jabs for en-AU and a strict, irony-centred one for en-UK (Appendix C), a choice made on OOF results and so not independent evidence. Whatever the cause, the contrast is consistent with the poor cross-variety transfer that BESSTIE reported. In our own experiment, a three-seed DeBERTa trained on one variety's pool folds and evaluated on the other's Reddit comments loses 0.14 to 0.16 sarcasm AUC (0.840 to 0.678 for en-AU, 0.859 to 0.715 for en-UK), while sentiment, evaluated on both sources, loses at most 0.011 AUC. Training on both varieties with the variety prefix does not hurt (0.845 and 0.861): a model told the variety can fit both label sets, which is what our per-variety outputs do.

6.3 How far can en-AU sarcasm go?

Agreement on en-AU sarcasm is low even between humans. BESSTIE reports a second annotator for 50 items per variety (Srirag et al., 2025, §2.3): $\kappa = 0.47$ for en-AU sarcasm, against 0.61 and 0.79 for sentiment; with 50 items, its uncertainty is large. Our submitted ensemble agrees with the gold en-AU sarcasm labels at $\kappa = 0.56$ (95% bootstrap interval 0.52–0.59) over all 2,004 en-AU Reddit comments. The samples differ, so the two numbers cannot be compared directly, and neither is a ceiling estimate; together they show that en-AU sarcasm labels are hard to reproduce, for humans as for models. The hardest slice is where §6.2 locates the difference between the varieties: among comments the annotator labelled negative, our ensemble's sarcasm AUC is 0.77 for en-AU against 0.91 for en-UK. Why these comments are hard, whether through the labels, missing context or model limits, remains open.

7 What Did Not Help

Once the three-model ensemble was in place, nothing we tried cleared the protocol's bar; Appendix A lists the main interventions, their baselines and their effects. Alternative backbones (ELECTRA, ModernBERT, an NLI-tuned DeBERTa) scored 0.009 to 0.016 below DeBERTa-v3-large and did not improve the ensemble. Training tweaks (weight averaging, layer-wise learning-rate decay), domain-adaptive pre-training and sarcasm pre-training on external data were ties or losses. A LoRA-tuned Qwen3.5-4B was weaker than the frozen probe and added nothing to the ensemble. Context did not help either: recovered thread text, the labels of other comments in the same thread, and a nearest-neighbour label vote moved the score by -0.005 to +0.001, within noise. The idea that looked best on a single fold, pseudo-labelling, failed its pre-specified confirmation on an untouched fold (-0.006).

8 Conclusion

A small ensemble, chosen under a strict protocol, shared first place. The more useful finding concerns the labels: the en-AU and en-UK sarcasm labels differ in rate and in the cues they follow, consistently across subreddits, and agreement on en-AU sarcasm is low even between humans. Cross-variety results on this benchmark therefore mix two things: how well a model understands each variety, and how differently each variety's sarcasm was labelled. Because the metric takes the weaker variety, en-AU sets the score in both tasks; en-AU sarcasm is the weakest component and the clearest priority, especially the negative comments where its labels are hardest to predict. We suggest that variety-sensitive benchmarks state explicitly whether mockery alone counts as sarcasm, have native annotators of each variety label a shared set of items, and report agreement per variety.

Limitations

BESSTIE's annotators could consider context we do not see. The exploratory LLM re-annotation (Appendix B) uses one model with a narrower guideline and is not evidence of annotation error. With one annotator per variety, each labelling different comments, we cannot separate individual, variety-level and text differences, so "the en-AU labels" should not be read as "Australian English".

Our comparison with BESSTIE's second annotator uses different samples. Our bootstrap intervals resample comments and do not model dependence between comments from the same thread. The cue analyses are exploratory, use automatic classifiers as proxies for derision and play, and do not correct for multiple comparisons. We did not test models larger than 9B parameters, so our curves do not rule out a capacity limit. Resampling pool predictions at test size gives a sampling standard deviation of about 0.013 on the official score, larger than most differences we report; we therefore lean on OOF estimates and paired resampling. The DeBERTa epoch count was fixed on the fold later used for screening, which makes those screens slightly optimistic. En-UK reviews are handled by a rule.

Ethics and Data

Our code is released at `https://github.com/Sekinal/alta2026-stargazers`, and the full release with predictions, cached features and models at `https://huggingface.co/Thermostatic/alta2026-thermostatic`. We redistribute no competition text: Table 1 is paraphrased, and the release excludes the data. To audit the data split we matched Reddit comments to a public archive; we stored only subreddit, thread and timing fields, no usernames, and release none of it. Competition texts were processed by Anthropic's models through Claude Code during development and by the blind readers, who received only the text, its variety and an opaque id; the GPT-6 Astra reviewers saw the paper and aggregate results, not the data. External data used in exploration came from publicly available sources under their own terms (Appendix D); none is redistributed.

Acknowledgements

All code, experiments and analyses, and the first draft of every section of this paper, were produced by Claude Code (Anthropic, 2026), running Claude Opus 5 and Claude Opus 5.5, under the author's direction; sub-agents of the same model also served as the blind readers (Appendix B) and as internal reviewers, and OpenAI's GPT-6 Astra, run through the Codex CLI, reviewed the paper independently in several rounds. The design drew on the author's experience in Kaggle competitions, including recipes adapted from the author's earlier AstroCLIMB system, and on a ChatGPT-generated list of LoRA variants with links to their papers that the author supplied; the agents contributed the ensemble, output-head, prefix and probe designs from their prior knowledge. The sarcasm and annotation literature was read after the system was fixed and did not change it. A full account of the development process, including mistakes found and corrected, is released with our code. The author reviewed and edited the paper and takes full responsibility for its content; AI systems are not authors of this work. We thank Lium for a generous grant of GPU credits that made this and other work possible, and the organisers, Diego Mollá-Aliod, Aditya Joshi and Dipankar Srirag, for running the shared task and for their careful significance analysis.

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A What Did Not Help: Full Results

Table 3 lists the main interventions tried after the three-model ensemble was fixed, under the protocol of §3. One-fold DeBERTa deltas compare three-seed prediction averages, LoRA deltas compare mean scores over two seeds, and five-fold deltas compare full OOF scores.

B Exploratory Check: Blind LLM Re-annotation

We asked whether a blind reader would reproduce the gold sarcasm labels. Two Claude Opus 5.5 sub-agents labelled 150 random Reddit comments per variety, plus about 50 per variety where our ensemble confidently contradicts the gold sarcasm label. Each saw only the text and its variety, in shuffled batches of 100 with opaque item ids, and this guideline: "You are annotating short Reddit comments from Australian (en-AU) or British (en-UK) English speakers. For each item give two labels, judging the text as written (you do not see the thread it came from): sarcasm: 1 if the author says something they do not literally mean in order to mock, criticise, or be ironic/teasing (sarcasm, verbal irony, mock praise, rhetorical ridicule); 0 otherwise. Plain complaints, jokes that are not ironic, and sincere statements are 0. sentiment: 1 if the overall attitude/emotion expressed is positive, 0 if it is negative (or neutral-leaning negative)." This guideline is narrower than BESSTIE's: it requires non-literal meaning and counts plain complaints as not sarcastic. The sample, guideline and criteria were written down before the readers were launched and committed to version control together with the results.

Intervention (protocol)	Δ
DeBERTa recipe / backbone (one fold, 3 paired seeds)	
weight EMA	+.001
layer-wise LR decay + EMA	+.004
NLI-tuned DeBERTa-v3-large	−.016
ELECTRA-large	−.009
ModernBERT-large	−.012
any of these inside the ensemble	−.007 to +.002
domain-adaptive pre-training	−.004
sarcasm pre-training (external data)	−.011
LLMs (Qwen3.5-4B Qwen Team, 2026, one fold)	
LoRA + rsLoRA, alone / in ensemble	−.02 / .000
EVA + rsLoRA	−.04
EVA + DoRA + rsLoRA + Riemannian	−.10
Data and context (five folds unless noted)	
pseudo-labels (pre-spec., fold 4)	−.006
recovered thread context (stacker)	−.001
same-thread label features	+.001
kNN label vote	−.005
LLM-reader label, en-UK (pre-spec.)	.000

Table 3: Change in official score against the matching baseline (DeBERTa rows and LoRA alone: three-seed DeBERTa on the same fold; LoRA variants: LoRA + rsLoRA; pseudo-labels: the unchanged ensemble on fold 4; other five-fold rows: the full ensemble, or for thread context the same stacker without context features). Methods: weight averaging (EMA); layer-wise learning-rate decay (Howard and Ruder, 2018); NLI-tuned DeBERTa (Laurer et al., 2024); ELECTRA (Clark et al., 2020); ModernBERT (Warner et al., 2025); domain-adaptive pre-training (Gururangan et al., 2020); sarcasm pre-training data (Appendix D); LoRA (Hu et al., 2022) with rsLoRA (Kalajdzievski, 2023), EVA (Paischer et al., 2025), DoRA (Liu et al., 2024) and Riemannian preconditioning (Zhang and Pilanci, 2024), via PEFT (Mangrulkar et al., 2022) (the EVA initialisation was capped at 32 batches and had not converged in 115 of 248 layers); pseudo-labelling (Lee, 2013), following a recipe from the author's earlier AstroCLIMB system. Only the alternative backbones, sarcasm pre-training, the LoRA variants and pseudo-labelling ($P_{\text{boot}} = 0.95$ for the baseline) reach $P_{\text{boot}} \geq 0.8$, all in the negative direction.

Results. The two readers agree with each other at $\kappa = 0.90$–0.98, so they behave like one reader; this shows repeatability, not validity. LLM labels can also differ systematically from human judgements on subjective tasks (Pavlovic and Poesio, 2024). On en-AU comments the gold labels call 43% sarcastic and the readers 16%; on en-UK comments, 29% and 27% (Table 4). Our ensemble agrees with the gold en-AU sarcasm labels ($\kappa = 0.56$) far more than with the readers (0.30–0.34).

	G–O	G–R	O–R	Positive rate G / O / R
Sarc AU	**.56**	.21–.28	.30–.34	.43 / .43 / .16
Sarc UK	.59	.62	.45	.29 / .23 / .27
Sent AU	.74	.60–.65	.60–.65	.29 / .33 / .26
Sent UK	.85	.81–.83	.67–.69	.17 / .15 / .22

Table 4: Agreement (Cohen's κ, Cohen, 1960) on 150 random Reddit comments per variety between the gold labels (G), our ensemble (O) and the two readers (R; ranges over readers), and the share each labels positive (sarcastic, or positive sentiment; R is the mean of the two readers). The 95% interval for G–O sarcasm is about ± 0.14.

Pre-specified criteria. We wrote down three criteria for a label-limited plateau: (i) the readers agree with each other more than with gold (held); (ii) our model agrees with the readers at least as much as with gold (failed in both varieties); (iii) on disputed items, agreeing readers side with our model more than 60% of the time (held: 31 to 14 en-AU, 31 to 17 en-UK). Criterion (iii) turned out to be weak: on en-AU comments the readers side with whichever label says "not sarcastic" (with our model 28 to 4 when it says not sarcastic, with gold 10 to 3 when gold does). On en-UK comments the pattern is less clear (26 to 16 and 5 to 1 for our model), with few items in the second cell. The pre-specified evidence therefore does not establish a label-limited plateau. Adding a reader's label to the ensemble as a pre-specified en-UK feature raised en-UK Reddit sarcasm AUC from 0.906 to 0.920 and en-UK sarcasm F1 from 0.841 to 0.843, but left the official score unchanged, because en-AU remained the binding variety.

Why this is not in the main text. The readers are one LLM, from the family that built our system, applying a stricter definition than BESSTIE's, and not native speakers of either variety. The low en-AU rate may reflect the definition, or a reader that misses Australian deadpan and jocular abuse, as much as the annotator. We report the study for completeness; the main analysis does not depend on it.

C Reproducibility Details

Folds and thresholds. Five folds over the pool (seed 2026), stratified by segment and both labels, with identical texts kept in one fold. Thresholds: one per task and segment, chosen by coordinate ascent to maximise the macro-F1 of the corresponding component; en-UK reviews are never sarcastic.

Reported OOF scores fit thresholds on four folds and apply them to the fifth; test thresholds are fitted on all OOF predictions.

Ensemble. For each text and task, the DeBERTa member's probability is the mean over its 25 models (5 seeds $\times$ 5 folds; OOF: the 5 seeds of the held-out fold); probe test probabilities average the five fold models. With $\ell(p) = \log \frac{p}{1-p}$, the ensemble probability is $\sigma\big(\frac{1}{3}[\ell(p_{\mathrm{DeB}}) + \ell(p_{\mathrm{e5}}) + \ell(p_{\mathrm{LLM}})]\big)$.

DeBERTa. First-token pooling, one sentiment logit and one sarcasm logit per variety (the row's variety selects the head); loss = binary cross-entropy for sentiment + binary cross-entropy for sarcasm, equal weights; AdamW (Loshchilov and Hutter, 2019), learning rate 10^{-5}, batch 16, 10% warmup, bf16, 4 epochs, maximum length 256 tokens (longer texts keep head and tail); seeds 13–17.

Embedding probe. e5-large-v2 embeddings with the prefix "query: " are concatenated with multilingual-e5-large-instruct embeddings computed with the instruction "Decide whether the given online review or comment is sarcastic or ironic"; features are standardised and a logistic regression is fitted per task, with C chosen by three-fold inner cross-validation.

LLM probe. Qwen3.5-9B in 4-bit NF4 (double quantisation, bf16 compute), chat template without thinking. Features: the hidden state at the last prompt position of layer 24, plus the logit margin between the answer words (case variants pooled by log-sum-exp); standardised; logistic regression ($C = 0.002$), with one sarcasm probe per variety. The prompt names the text type (e.g. "a Reddit comment from an Australian community (Australian English)"), gives a definition and asks "Is the text sarcastic? Answer Yes or No." The en-AU definition reads: "Sarcasm here is understood broadly: the writer uses irony, mockery, ridicule, cynical or derisive jabs, snark, or says something they do not literally mean, usually to criticise or show contempt (for example cynical remarks about politicians, parties or government). A sincere, neutral, informative or friendly text is NOT sarcastic." The en-UK definition reads: "Sarcasm means the writer says something they do not literally mean, usually the opposite, in order to mock, criticise or show contempt. [...] A text that is merely angry, negative, rude or critical but says exactly what it means is NOT sarcastic. British sarcasm is often dry and understated [...]"; the elided parts list typical forms and set phrases. The sentiment prompt asks whether the overall sentiment is positive or negative and states that sarcastic praise counts as negative. Full prompts are in the release.

Cue scores. Each NLI cue is the mean entailment log-odds of `MoritzLaurer/deberta-v3-large-zeroshot-v2.0` over a few hypotheses, with the comment as premise (256 tokens, head and tail): *mocking*: "The author is mocking someone.", "The author is ridiculing something.", "This text makes fun of someone."; *contempt*: "The author is contemptuous.", "The author looks down on someone."; *anger*: "The author is angry.", "The author is frustrated.", "The author is annoyed."; *cynicism*: "The author is cynical.", "The author expects things to go badly."; *playful joking*: "The author is joking.", "This text is meant to be funny.", "The author is being playful."; *praise*: "The author is praising something.", "The author is happy with something.", "This is a compliment."; *ironic praise*: "This is ironic praise.", "The author pretends to be positive but is actually negative.", "The compliment in this text is not meant seriously." *Joy* is the emotion classifier's joy probability. The other NLI concepts computed (opposite meaning, sincerity, complaint, politics, exaggeration, rhetorical question, positive, negative, advice) were not used as cues; we chose the eight shown as the most interpretable markers of derision and play. AUC intervals resample comments (1,000 replicates).

D Resources

Final system. `microsoft/deberta-v3-large` (He et al., 2023); `intfloat/e5-large-v2` (Wang et al., 2022); `intfloat/multilingual-e5-large-instruct` (Wang et al., 2024); `Qwen/Qwen3.5-9B` (Qwen Team, 2026) with bitsandbytes 4-bit (NF4) quantisation (Dettmers et al., 2023); PyTorch (Paszke et al., 2019), Transformers (Wolf et al., 2020), sentence-transformers (Reimers and Gurevych, 2019) and scikit-learn (Pedregosa et al., 2011); data processing with Polars (Vink et al., 2024) and figures with Matplotlib (Hunter, 2007). Trained only on the shared-task training and development data. Checkpoints screened in Table 3: `google/electra-large-discriminator`, `answerdotai/ModernBERT-large`, `MoritzLaurer/DeBERTa-v3-large-`

`mnli-fever-anli-ling-wanli` and `Qwen/Qwen3.5-4B`.

Analyses. `MoritzLaurer/deberta-v3-large-zeroshot-v2.0` (Laurer et al., 2023), `siebert/sentiment-roberta-large-english` (Hartmann et al., 2023) and `j-hartmann/emotion-english-distilroberta-base` (Hartmann, 2022) for the cues of Figure 4; Reddit thread metadata matched through Arctic Shift (Heitmann, 2024) for the split audit.

External data in the reported negative results. The sarcasm pre-training of Table 3 used Reddit comments tagged `/s` (tag removed) from about 25 AU and 25 UK subreddits, collected through Arctic Shift, ConvoKit's per-subreddit corpora (Chang et al., 2020) and the UK subreddit dumps of alujjdnd (2023), with untagged comments from the same subreddits as negatives; it added EPIC (Frenda et al., 2023), iSarcasmEval (Abu Farha et al., 2022), SemEval-2018 Task 3 (Van Hee et al., 2018), FigLang 2020 (Ghosh et al., 2020) and Wallace et al. (2014). Domain-adaptive pre-training and pseudo-labelling used unlabelled in-domain text: AU and UK Reddit via Arctic Shift (Reddit's terms), Google Local reviews (He et al., 2017; Pasricha and McAuley, 2018), Booking.com reviews (`huggingface.co/datasets/Booking-com/accommodation-reviews`, CC BY-SA 4.0), NHS GP reviews (`huggingface.co/datasets/janduplessis886/england-nhs-gp-reviews`, Open Government Licence) and Inside Airbnb reviews (`insideairbnb.com`, CC BY 4.0). All external data were deduplicated against the task data, and none is used by the final system.

Other exploration. We also screened other encoders, embedding models, LLMs, zero-shot feature models and classical models (TF-IDF, SVMs, gradient-boosted trees), and collected further sarcasm and sentiment corpora. None contributed to the final system or to any result reported here; the release lists them all with their sources.